

Preliminary Numerical Analysis: Convergence and Error Analysis of Implicit and Crank-Nicolson Finite Difference Schemes with Payoff Smoothing

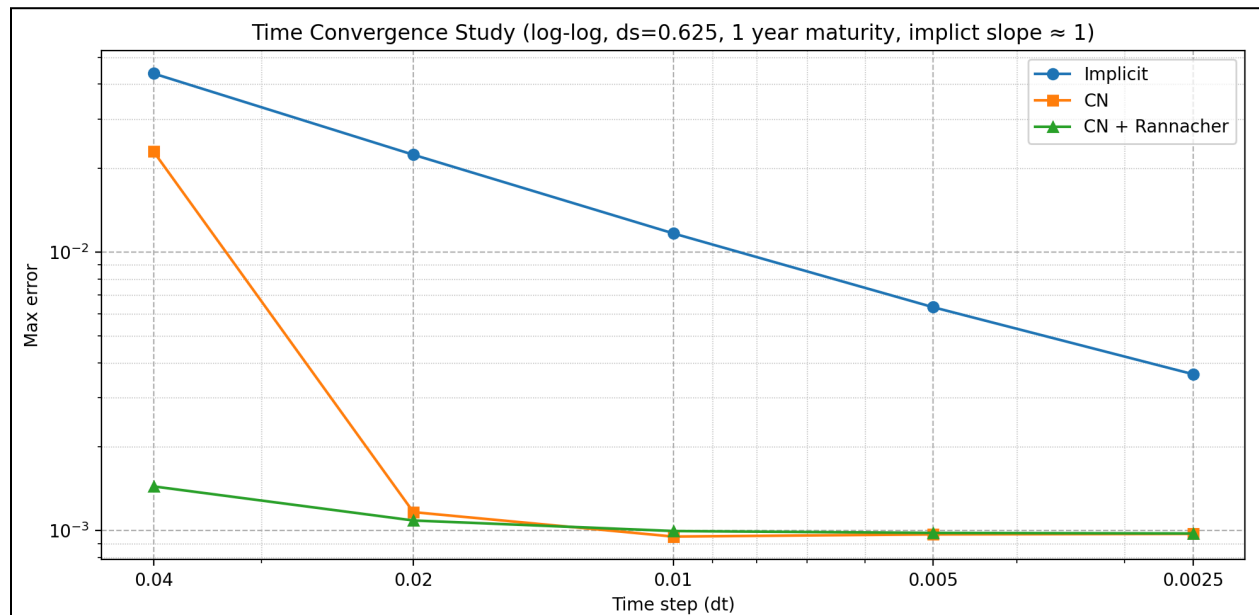
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Github: <https://github.com/utsochaudhuri/BSM-PDE-Numerical-Methods>

Introduction

This report presents preliminary numerical analysis work demonstrating the implementation and convergence behaviour of several finite difference schemes for pricing European options, compared against the Black Scholes Merton (BSM) solution. The methods examined span three broad categories: the implicit method, the Crank-Nicolson (CN) method, and Crank-Nicolson with Rannacher time-stepping. Within the CN framework, the effect of three payoff smoothing approaches: no smoothing, cell average smoothing, and quadratic smoothing are investigated, yielding a total of nine method configuration combinations (implicit also included). Both temporal and spatial convergence are studied. The primary metric used throughout in this study is the maximum absolute error relative to the BSM price across all stock price nodes.

This report uses graphs to present the most relevant results.



Plot 1

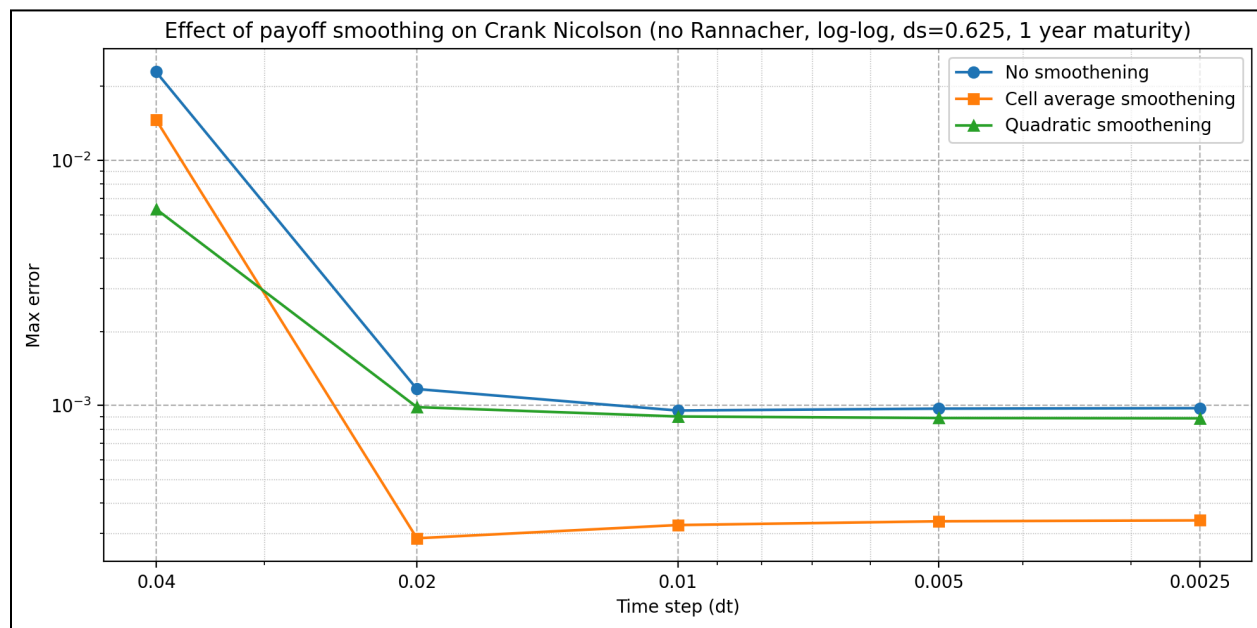
Plot 1 shows a log-log time convergence study, comparing the three core methods: Implicit, CN, and CN + Rannacher, with a fixed spatial step of $ds = 0.625$ and a 1-year maturity.

The implicit method follows a clean, straight-line decay on the log-log scale, confirming first-order convergence in time (slope ≈ 1), which lines up with its theoretical $O(dt)$ accuracy. Its errors are noticeably higher than the CN-based methods across all time steps tested, this tradeoff is associated with the implicit scheme: it is unconditionally stable but only first-order accurate.

CN without Rannacher behaves less predictably at coarser time steps, with a sharp drop in error from $dt = 0.04$ to $dt = 0.02$ before settling and tracking close to CN + Rannacher. This kind of behaviour is a well-known consequence of the non-smooth payoff discontinuity at the strike interacting poorly with the CN scheme when time steps are large, and it is precisely what Rannacher time-stepping is designed to address.

CN + Rannacher starts at a lower error even at the coarsest time step and decreases steadily throughout. This shows that replacing the first few CN steps with fully implicit steps is enough to suppress the oscillations caused by the payoff kink, allowing the scheme to recover its second-order convergence cleanly from the start.

It is worth noting that all three methods level off at around 10^{-3} as the time step gets finer. At this point, reducing dt further does not help much because the spatial discretisation error (from the fixed $ds = 0.625$ grid) has become the dominant source of error.



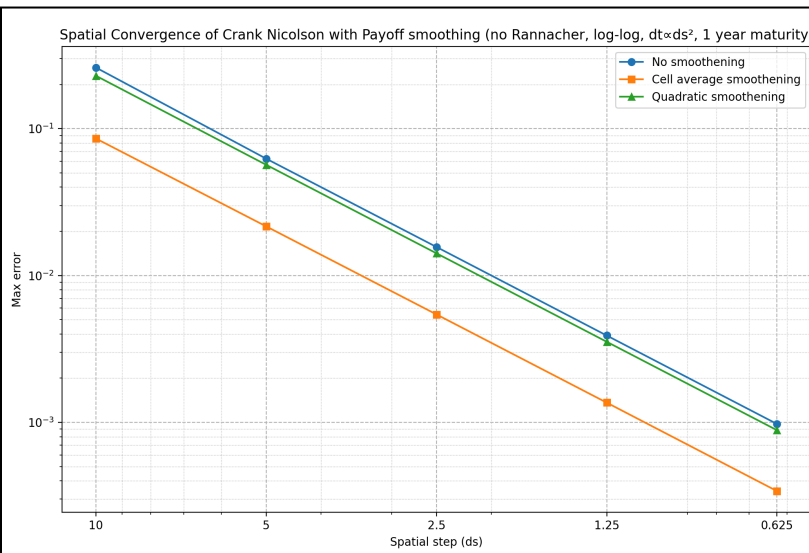
Plot 2

Plot 2 isolates the effect of payoff smoothing techniques on the Crank-Nicolson scheme without Rannacher time-stepping, comparing no smoothing, cell average smoothing, and quadratic smoothing across the same time step range.

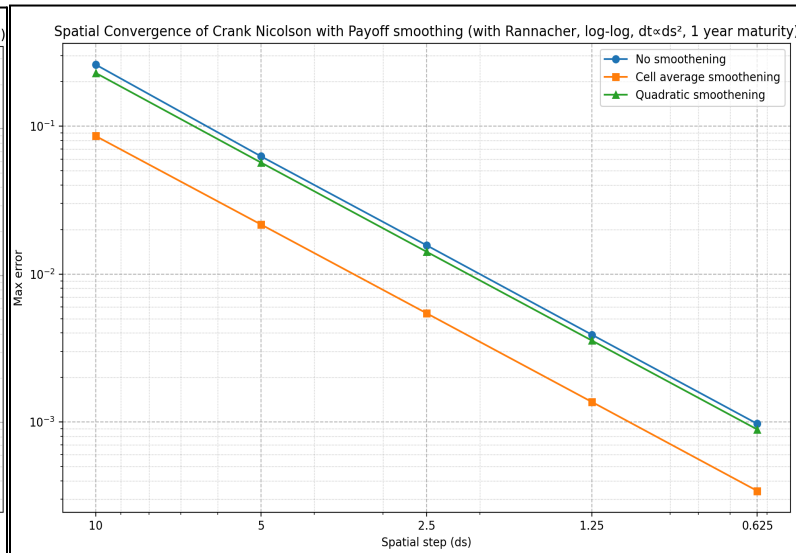
Without any smoothing, CN converges relatively slowly and from a high starting error at $dt = 0.04$, consistent with the observations in Plot 1. The errors decrease steadily but plateau around 10^{-3} , with no strong distinction from quadratic smoothing at finer time steps.

Quadratic smoothing provides a modest improvement over no smoothing, particularly at coarser time steps, though the two methods converge to nearly the same error floor at fine resolutions. This suggests quadratic smoothing helps regularise the payoff at the strike sufficiently to reduce early-time errors, but its advantage diminishes as dt decreases.

Cell average smoothing stands out as the most effective among the three methods. It achieves a substantially lower error across all time steps and converges to a floor roughly half an order of magnitude below the other two methods. The improvement is persistent and does not just manifest at coarse time steps, suggesting that cell average smoothing more accurately represents the discontinuity in the payoff by distributing it across a grid cell in a manner that is more compatible with the finite difference stencil.



Plot 3a



Plot 3b

These two plots examine the spatial convergence of the Crank-Nicolson scheme with the three payoff smoothing options, using a coupled refinement strategy where $dt \propto ds^2$ to ensure that temporal errors do not dominate. Plot 3a uses CN without Rannacher and plot 3b uses CN with Rannacher. The two plots are presented together as their structure is nearly identical, making a direct comparison between them straightforward.

In both plots, CN without smoothing and CN with quadratic smoothing trace almost identical convergence paths, both exhibiting straight-line decay on the log-log scale corresponding to second-order spatial convergence (slope ≈ 2). This is the expected theoretical rate for CN under smooth conditions, and the tight overlap of these two lines confirms that quadratic smoothing does not alter the spatial convergence order.

Cell average smoothing, however, consistently achieves errors approximately one order of magnitude lower than the other two approaches across all spatial refinement levels, while maintaining the same second-order convergence rate. This confirms that cell average smoothing reduces the absolute error beforehand rather than changing the convergence order, which is a meaningful practical advantage, as it translates directly to higher accuracy for any given grid resolution.

Comparing the two plots, the addition of Rannacher time-stepping produces no significant change in the spatial convergence behaviour. The curves in both plots are nearly indistinguishable, which is expected: Rannacher smoothing acts on the temporal component of the error, and since $dt \propto ds^2$ in this study, the temporal errors are already well-controlled. This confirms that the benefits of Rannacher are primarily temporal in nature and do not interfere with spatial accuracy.

Conclusion

Across all four plots, a coherent picture emerges. The implicit scheme is reliable and first-order accurate but consistently less accurate than the CN-based methods. CN without Rannacher achieves second-order accuracy in principle but is sensitive to the non-smooth payoff at coarser time steps, exhibiting erratic convergence that Rannacher time-stepping corrects efficiently. Of the payoff smoothing techniques, cell average smoothing stands out as the most impactful, consistently reducing the absolute max error by roughly a factor of 5–10 compared to no smoothing or quadratic smoothing, while preserving the second-order convergence rate in space. Quadratic smoothing offers marginal improvement over no smoothing, particularly in time convergence studies. Finally, the spatial convergence analysis confirms that Rannacher and payoff smoothing address distinct sources of error and can be combined without adverse interaction, providing a robust and accurate overall scheme for option pricing under the BSM framework.